

Jacob A. Parker, B.Sc.

Ph.D. Candidate in Neuroscience

Neuroscience Graduate Group

University of Pennsylvania

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EDUCATION

- 2021–Present **Ph.D. in Neuroscience [in progress]**
University of Pennsylvania
Thesis Advisors: Joshua I. Gold & Joseph W. Kable
- 2013–2017 **B.Sc. in Neural Science**
New York University
Honors: Magna Cum Laude

POSITIONS

- 2017–2021 **Post-Baccalaureate Research Fellow**
Human Motor Control Section
National Institute of Neurological Disorders and Stroke
Principal Investigator: Mark Hallett
- 2015–2017 **Undergraduate Research Assistant**
Carrasco Lab
New York University
Supervisors: Rachel Denison & Marisa Carrasco

HONORS & AWARDS

- 2023–2026 NSF Graduate Research Fellowship
- 2025 Jameson-Hurvich Travel Award, Penn
- 2017–2020 NIH Intramural Research Training Award
- 2017 NIH Summer Internship Program
- 2017 University Honors Scholar, NYU
- 2014–2017 Dean's List, NYU
- 2014–2016 CSCAA Honorable Mention Scholar All American (academic + athletic excellence)
- 2014–2016 University Athletic Association All Academic Recognition

PUBLICATIONS

- [7] **Jacob A. Parker**, Alexandre L. S. Filipowicz, Kristen Li, Joseph W. Kable, Joshua I. Gold. Suboptimal human decision-making reflects an efficient information bottleneck on inference. *[In preparation]*.
- [6] Primavera A. Spagnolo, **Jacob A. Parker**, Mark Hallett, Silvina G. Horovitz. Functional Movement Disorder Is Associated with Abnormal Interoceptive Brain Activity: A Task-Based Functional MRI Study. *Frontiers in Psychiatry* **16** (2025).
- [5] Rebecca E. Waugh, **Jacob A. Parker**, Mark Hallett, Silvina G. Horovitz. Classification of Functional Movement Disorders with Resting-State Functional Magnetic Resonance Imaging. *Brain Connectivity* **13**, 4–14 (2023).
- [4] **Jacob A. Parker**, Shabbir H. Merchant, Sanaz Attaripour-Isfahani, Hyun Joo Cho, Patrick McGurrin, Brian P. Brooks, Albert R. La Spada, Mark Hallett, Laryssa A. Huryn, Silvina G. Horovitz. In Vivo Assessment of Neurodegeneration in Spinocerebellar Ataxia Type 7. *NeuroImage: Clinical* **29**, 102561 (2021).
- [3] Primavera A. Spagnolo, **Jacob Parker**, Silvina Horovitz, Mark Hallett. Corticolimbic Modulation via Intermittent Theta Burst Stimulation as a Novel Treatment for Functional Movement Disorder: A Proof-of-Concept Study. *Brain Sciences* **11**, 791 (6 2021).
- [2] Shabbir Hussain I. Merchant, Eleni Frangos, **Jacob Parker**, Megan Bradson, Tianxia Wu, Felipe Vial-Undurraga, Giorgio Leodori, M. C. Bushnell, Silvina G. Horovitz, Mark Hallett, Traian Popa. The Role of the Inferior Parietal Lobule in Writer's Cramp. *Brain* **143**, 1766–1779 (2020).
- [1] Rachel N. Denison*, Jacob A. Parker*, Marisa Carrasco. Modeling Pupil Responses to Rapid Sequential Events. *Behavior Research Methods* (2020). *equal contribution.

CONFERENCE PRESENTATIONS

- [15] **J. A. Parker**, A. Filipowicz, K. Li, V. Balasubramanian, J. W. Kable, J. I. Gold. *Sub-optimal human decision-making reflects an efficient information bottleneck on inference*. Cognitive Computational Neuroscience 2025. Amsterdam, Netherlands, 2025. [Poster].
- [14] M. C. Tandoc, S. H. Solomon, A. B. Williams, A. Gordienko, **J. A. Parker**, A. C. Schapiro. *Category learning drives neural repulsion initially but integration at a delay*. Cognitive Neuroscience Society Annual Meeting. Boston, MA, USA, 2025. [Poster].
- [13] **J. A. Parker**, A. Filipowicz, K. Li, V. Balasubramanian, J. W. Kable, J. I. Gold. *Sub-optimal human decision-making can reflect an efficient information bottleneck*. Society for Industrial and Applied Mathematics Conference on the Life Sciences. Portland, OR, USA, 2024. [Invited Talk].

- [12] M. C. Tandoc, S. H. Solomon, **J. A. Parker**, A. Gordienko, A. C. Schapiro. *The representation and retrieval of general versus specific category knowledge*. Cognitive Neuroscience Society Annual Meeting. Toronto, Canada, 2024. [Poster].
- [11] **J. A. Parker**, A. Filipowicz, V. Balasubramanian, J. W. Kable, J. I. Gold. *Suboptimal human decision-making can reflect an efficient information bottleneck*. Society for Neuroscience Annual Meeting. Washington, DC, USA, 2023. [Poster].
- [10] S. G. Horovitz, **J. A. Parker**, P. Bedard, C. Maurer, M Hallett. *Structural alterations in functional movement disorders: a diffusion weighted imaging study*. 28th Annual Meeting of the International Society for Magnetic Resonance in Medicine. Virtual, 2020. [Poster].
- [9] **J. A. Parker**, Shabbir H. Merchant, Sanaz Attaripour-Isfahani, Patrick McGurrin, Laryssa A. Huryn, Mark Hallett, Silvina G. Horovitz. *Structural Magnetic Resonance Imaging Correlates of Neurodegeneration in a Spinocerebellar Ataxia Type 7 Cohort*. Society for Neuroscience Annual Meeting. Chicago, IL, USA, 2019. [Poster].
- [8] S. H. Merchant, E. Frangos, S. G. Horovitz, T. Popa, **J. A. Parker**, M. Hallett. *Interactions within Fine Motor Control Network in Task-specific Dystonia*. International Congress of Parkinson’s Disease and Movement Disorders. Nice, France, 2019. [Poster].
- [7] R. N. Denison, **J. A. Parker**, M. Carrasco. *Estimation of pupillary responses to rapid events*. 9th Annual Meeting of the Vision Sciences Society. St. Pete Beach, FL, USA, 2019. [Poster].
- [6] **J. A. Parker**, S. H. Merchant, S. Attaripour-Isfahani, P. McGurrin, L. A. Huryn, M. Hallett, S. G. Horovitz. *Structural Magnetic Resonance Imaging Correlates of Neurodegeneration in a Spinocerebellar Ataxia Type 7 Cohort*. American Academy of Neurology Annual Meeting. Philadelphia, PA, USA, 2019. [Poster].
- [5] S. Attaripour-Isfahani, P. McGurrin, **J. A. Parker**, N. Dang, F. Vial, M. Hallett. *Deciding and Instructed to go and no-go*. Second International Conference on Neuroscience and Free Will. Orange, CA, USA, 2019. [Poster].
- [4] **J. A. Parker**, S. H. Merchant, S. Attaripour-Isfahani, P. McGurrin, M. Hallett, S. G. Horovitz. *Pathophysiologic insights into Spinocerebellar Ataxia Type 7 through Structural Imaging Analysis*. Society for Neuroscience Annual Meeting. San Diego, CA, USA, 2018. [Poster].
- [3] **J. A. Parker**, S. H. Merchant, S. Attaripour-Isfahani, P. McGurrin, M. Hallett, S. G. Horovitz. *Pathophysiologic insights into ataxia and spasticity through structural imaging of a Spinocerebellar Ataxia Type 7 cohort*. International Congress of Parkinson’s Disease and Movement Disorders. Hong Kong, China, 2018. [Poster].
- [2] R. N. Denison, **J. A. Parker**, M. Carrasco. *Pupil dilation reveals the timecourse of voluntary temporal attention*. 40th European Conference on Visual Perception. Berlin, Germany, 2017. [Poster].
- [1] **J. A. Parker**, R. N. Denison, M. Carrasco. *Pupil Size as an Index of Voluntary Temporal Attention*. 43rd Undergraduate Research Conference of the New York University College of Arts and Science. New York, NY, USA, 2017. [Talk].

MENTORSHIP

- 2024-Present **Leo Li**, Masters Student
- 2023-Present **Kristen Li**, Undergraduate Independent Study
Honors Thesis: A study on information-efficient inference in human decision-making
Received the Rose Undergraduate Research Award (only 10 undergrads awarded per year)

TEACHING

- Spring 2024 **Teaching Assistant**, Theoretical and Computational Neuroscience
University of Pennsylvania

SERVICE

- 2025-Present **Committee Member**, Penn Computational Neuroscience Initiative Seminar Series
- 2021-Present **General Member**, Graduate Led Initiatives and Activities (GLIA), UPenn
- 2022-2024 **Co-Chair**, Penn Neuroscience Public Lecture Series
Organized semi-annual lectures attended by 100+ people with annual budget of \$7000.
- 2021-2022 **Committee Member**, Penn Neuroscience Public Lecture Series
- 2019-2020 **Co-Chair**, NIH Postbac Seminar Series
- 2019-2020 **Volunteer Tutor**, STEM in Your Hood, College Bound DC
- 2018-2020 **Committee Member**, NIH Postbac Committee
- 2018-2020 **Patient Care Assistant**, Suburban Hospital (Bethesda, MD)
- 2018-2019 **Committee Member**, NIH Postbac Seminar Series
- 2016-2017 **Patient Care Volunteer**, New York Presbyterian-Brooklyn Methodist Hospital

SOFTWARE TOOLS

Pupil Response Estimation Toolbox
Website: <https://github.com/jacobaparker/PRET>
Publication: <https://pubmed.ncbi.nlm.nih.gov/32144729/>

ADDITIONAL EXPERIENCE

- Summer 2025 **Volunteer Teacher**, Penn Upward Bound, UPenn GLIA
Created and taught a lesson on neural circuits for a Neuroscience elective class offered as part of Upward Bound, a college prep program funded by the US Dept. of Education.

- 2023-2024 **Writer**, PennNeuroKnow, UPenn GLIA
 Authored 4 original articles about neuroscience for a non-scientific audience.
- Fall 2023 **Team Member - Healthcare Consulting Project**, Penn Biotech Group
 Built an AI-based system (GPT4) to extract structured data directly from the raw text of primary research articles for a biotech company specializing in cellular therapies.
Received Best Team Member Award
- 2022 **Team Member - Memo Writing Group**, Penn Science Policy and Diplomacy Group
 Co-authored a science policy memo detailing how social media algorithms facilitate the spread of disinformation and potential policy options to address the issue.
 Met with state and federal legislators to present and discuss content of memo.

ADDITIONAL WRITINGS

- [5] **Jacob A. Parker.** *Decide Quickly or Decide Accurately- How Your Brain Solves a Classic Decision-Making Problem.* PennNeuroKnow. (2024).
- [4] **Jacob A. Parker.** *The Experiments That Opened the Brain's Black Box.* PennNeuroKnow. (2024).
- [3] **Jacob A. Parker.** *Why Forgetting Might Actually Be a Good Thing.* PennNeuroKnow. (2024).
- [2] **Jacob A. Parker.** *ChatGPT versus the Brain: A Nerve-Wracking Matchup.* PennNeuroKnow. (2023).
- [1] Walker Gosrich, **Jacob A. Parker**, Hersh Sangvhi, Tuhina Srivastava, Shannon L. Wolfman. *Neutralizing the Algorithm: Approaches for Reducing the Spread of Disinformation Online.* SciTech Forefront. (2022).